

Source Attribution and Risk Assessment of Antimicrobial Resistance

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ABSTRACT Source attribution and microbial risk assessment methods have been widely applied for the control of several foodborne pathogens worldwide by identifying (i) the most important pathogen sources and (ii) the risk represented by specific foods and the critical points in these foods' production chains for microbial control. Such evidence has proved crucial for risk managers to identify and prioritize effective food safety and public health strategies. In the context of antimicrobial resistance (AMR) from livestock and pets, the utility of these methods is recognized, but a number of challenges have largely prevented their application and routine use. One key challenge has been to define the hazard in question: Is it the antimicrobial drug use in animals, the antimicrobial-resistant bacteria in animals and foods, or the antimicrobial resistance genes that can be transferred between commensal and pathogenic bacteria in the animal or human gut or in the environment? Other important limitations include the lack of occurrence and transmission data and the lack of evidence to inform dose-response relationships. We present the main principles, available methods, strengths, and weaknesses of source attribution and risk assessment methods, discuss their utility to identify sources and estimate risks of AMR from livestock and pets, and provide an overview of conducted studies. In addition, we discuss remaining challenges and current and future opportunities to improve methods and knowledge of the sources and transmission routes of AMR from animals through food, direct contact, or the environment, including improvements in surveillance and developments in genotypic typing methods.

INTRODUCTION

Antimicrobial use in humans and animals has been identified as a main driver of antimicrobial resistance (AMR), and bacteria harboring resistance to antimicrobials can

be found in humans, animals, foods, and the environment. As a consequence, humans can be exposed to antimicrobial-resistant bacteria through a wide range of sources and transmission pathways. To inform policies aimed at reducing the burden of AMR in animals and foods, risk managers need evidence about the most important sources and transmission routes and the critical points throughout the production chain for the prevention and control of AMR. While this process is complex and deeply reliant on the integration of surveillance data from humans, animals, and foods, it is supported by scientific disciplines that have evolved rapidly in the past decades, including source attribution and quantitative risk assessment.

Source attribution is a relatively new discipline that has been developed to assist risk managers to identify and prioritize effective food safety intervention measures. It is defined as the *partitioning of the human disease burden of one or more foodborne illnesses to specific sources, where the term source includes reservoirs and*

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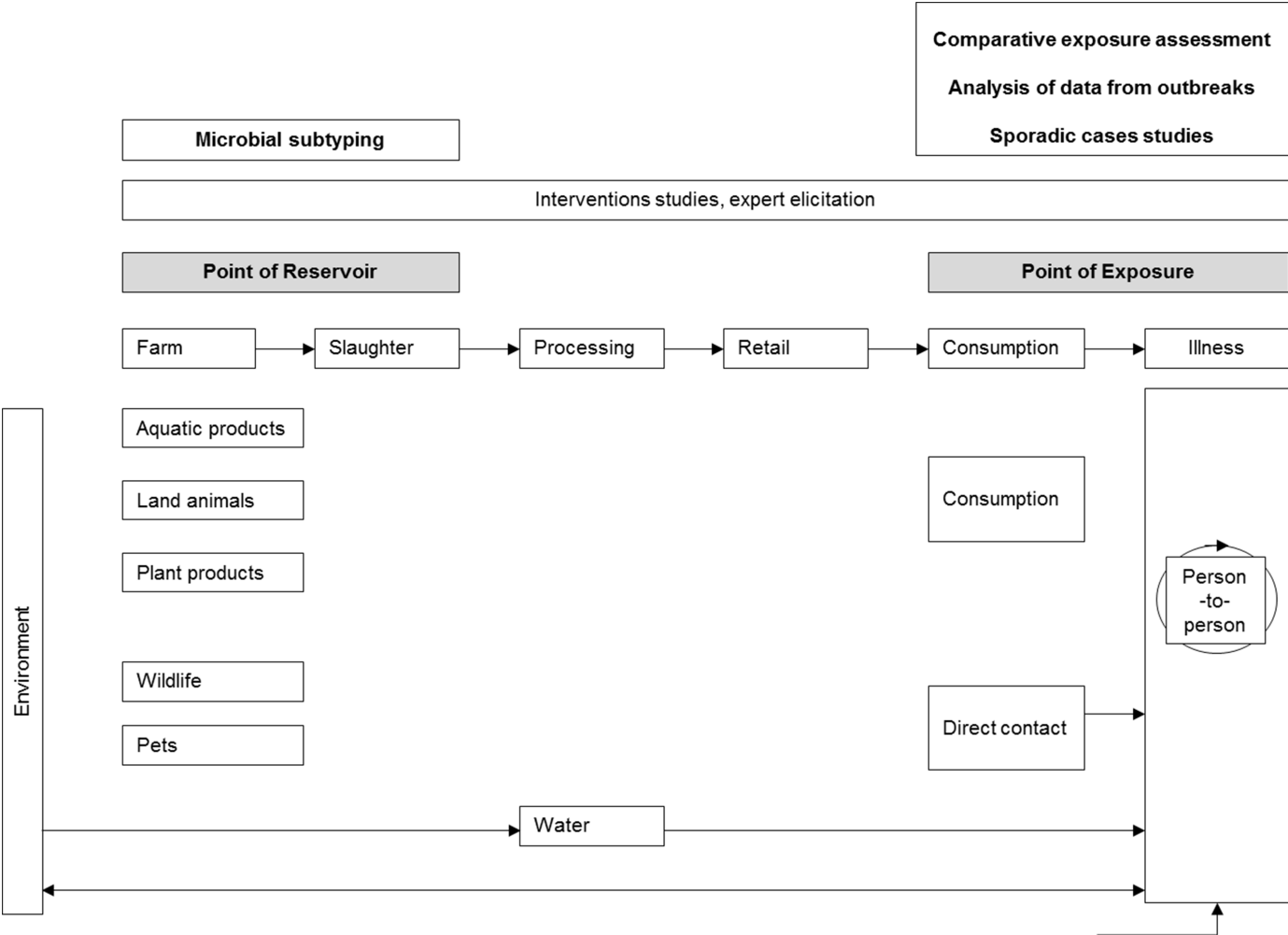
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vehicles (1). A variety of source attribution methods is available to estimate the relative contribution of different reservoirs or vehicles of foodborne pathogens, including methods relying on data on the occurrence of the pathogen in sources and humans, epidemiological studies, intervention studies, and expert elicitations. These methods have been applied to inform food safety policy-making at the national or international level, particularly regarding *Salmonella* and *Campylobacter* intervention strategies (see, e.g., 2–6). Source attribution methods differ in their approaches and data requirements, and as a consequence, they attribute disease at different points along the food chain (points of attribution), i.e., at the point of reservoir (e.g., animal production stage, environment emissions) or the point of exposure (end of the transmission chain) (Fig. 1). The application and utility of each method, therefore, de-

pends on the risk management question being addressed and on the availability of data.

Microbial risk assessment is a systematic and science-based approach to estimating the risk of microbial hazards in the production-to-consumption chain (8, 9). Microbial risk assessment can be used to detect critical control points along the food chain and for the assessment of control and intervention strategies. It is a well-established discipline that has been widely applied to estimate the risk of an extensive variety of pathogen-food commodity pairs, and it is also systematically applied to inform food safety risk management in many countries and international bodies such as the European Food Safety Authority (e.g., 10–12). In coordination with source attribution studies, it is particularly useful to focus on the production chain of the most important source(s) of the hazard of interest (as identified in

FIGURE 1 Routes of transmission of zoonotic pathogens and points of source attribution. Adapted from reference 7.



the source attribution step), identify the steps in the food chain that are critical for hazard control, and identify and suggest strategies for reduction of the risk to humans.

While source attribution and risk assessment have been widely used to provide evidence that can support strategies to reduce the burden of a number of foodborne pathogens, the transmission and spread of pathogens carrying resistance to antimicrobials adds an extra layer of complexity to this integrated food safety paradigm. On one hand, virtually any foodborne pathogen can acquire resistance to antimicrobials, which may lead to prolonged and more severe disease and even be life-threatening, when antimicrobial therapy is required but fails to succeed due to resistance to the prescribed drug(s). On the other hand, the potential transfer of AMR genes (i.e., the gene[s] carrying the resistance trait) between pathogenic and commensal bacteria in the human gut can amplify the public health impact of foodborne AMR (13). As a consequence, it is not only challenging to estimate the direct risk posed by resistant foodborne pathogens, but also to quantify the relative contribution to risk of the transfer of AMR genes, e.g., from commensals originating from animal reservoirs to human pathogens.

This article describes the overall concepts and methods in source attribution and microbial risk assessment, describes the state of the art of their application in the area of AMR, and discusses current challenges and future perspectives for the development of methods to inform policies to reduce the disease burden of AMR in human populations.

SOURCE ATTRIBUTION

Source Attribution of AMR

The purpose of applying source attribution methods to antimicrobial-resistant pathogens (i.e., a pathogen that has acquired resistance to at least one antimicrobial drug) or AMR genes is to identify the most important sources and transmission routes for human exposure to AMR. It is widely recognized that one of the main drivers of resistance in zoonotic bacteria is antimicrobial use in livestock production (i.e., in the reservoirs) (14). Identifying the most important reservoirs for human exposure to AMR is hence critical to direct policy-making aimed at reducing antimicrobial use at the primary production level. In addition, knowledge of the transmission routes from reservoirs to humans is crucial for the prioritization of risk management along the food chain.

While a range of source attribution methods attributing disease to the original reservoirs or to exposure

routes of foodborne pathogens exists, only a few studies have applied these in the context of AMR, and the relative importance of transmission pathways of resistance remains a critical knowledge gap.

The challenges of applying source attribution methods for AMR include the fact that virtually any pathogen can become resistant to antimicrobials and that most zoonotic pathogens can be transmitted to humans via a variety of foodborne and nonfoodborne routes. Thus far, source attribution has typically focused on a single pathogen (e.g., *Salmonella* or *Escherichia coli*) and on resistance profiles found among that pathogen in different sources (15–17). In addition, AMR genes are often located on plasmids, which can be transferred between bacterial species (plasmid-mediated horizontal gene transfer) and therefore also from commensal bacteria to human pathogens (e.g., *Klebsiella* spp.). Focusing on a single bacterial species is therefore likely to underestimate the overall exposure and thus the risk posed by AMR.

To address this challenge, source attribution of the AMR determinant may be more efficient. Such studies require knowledge of and data on the prevalence, abundance, and transmission of genes and on horizontal gene transfer rates, which is still being gathered (e.g., in the European Union project EFFORT—Ecology from Farm to Fork of Microbial Drug Resistance and Transmission; <http://www.effort-against-amr.eu/>).

Existing Source Attribution Approaches

Microbial subtyping

The microbial subtyping approach involves characterization of the hazard by subtyping methods (e.g., phenotypic or genotypic subtyping of bacterial strains), and the principle is to compare the subtypes of isolates from different sources (e.g., animals, food) with the subtypes isolated from humans. The subtyping approach attributes illness at the point of reservoir and is enabled by the identification of strong associations between some of the dominant subtypes and a specific reservoir or source, providing a heterogeneous distribution of subtypes among the sources (1).

Microbial subtyping methods for source attribution include frequency-matched models and population genetic models. While the frequency-matched methods are based on the comparison of human strain types and the distribution of those types in the sources, the population genetic models are based on modeling the organism's evolutionary history (18). In the frequency-matched models, subtypes exclusively or almost exclusively isolated from one source are regarded as indicators

for the human health impact of that particular source, assuming that all human cases caused by these subtypes originate only from that source. Human cases of disease caused by subtypes found in several reservoirs are then distributed relative to the prevalence of the indicator types (2, 3, 19). Population genetics approaches use genotyping data to infer evolutionary and clonal relationships among different strains, including the occurrence of novel (combinations of) alleles in strains from humans that are unobserved in source populations (20).

All microbial subtyping models require a collection of temporally and spatially related isolates from various sources and thus are facilitated by an integrated foodborne disease surveillance program providing a collection of isolates from the major animal reservoirs of foodborne diseases. These models do not require prevalence data and can rely on the distribution of the isolates' subtypes in the different sources and in humans.

Both types of models have been applied to attribute foodborne pathogens to sources in a variety of countries. Microbial subtyping approaches have been particularly successful to attribute *Salmonella* and *Campylobacter* infections (see e.g., 3, 21–24). This method has also been applied to other pathogens (namely, *Listeria monocytogenes* and Shiga toxin-producing *E. coli* [25, 26]), even though less frequently due to a lack of available surveillance data in most countries.

The microbial subtyping approach has seldom been used to estimate the relative contribution of sources of antimicrobial-resistant pathogens to AMR in humans. To our knowledge, two frequency-matched studies have been conducted, both using antimicrobial susceptibility patterns as a typing method for *Salmonella* (15, 16). Both studies demonstrate that AMR data can be used to characterize pathogen subtypes in a microbial subtyping source-attribution model and discuss its utility in terms of discriminatory power but do not focus on the source origin of specific AMR genes.

Microbial subtyping methods are recognized as one of the most robust types of data-driven methods for source attribution. They have the advantage of attributing illness to the reservoirs of the pathogens, thus informing risk-management strategies that are the closest possible to the original sources and preventing further spread to other routes or sources of transmission (1). Another advantage of this approach is that it does not require data on the prevalence and concentration of the pathogen in the different sources (which is often difficult to obtain) or on the exposure frequency in the population. Still, these methods are often limited by the requirement of comparable subtyping data originating

from an operative integrated surveillance of human cases and food/animals. In addition, the methods cannot distinguish between different transmission routes from a specific animal reservoir to humans.

Comparative exposure assessment

Comparative exposure assessments determine the relative importance of the known transmission routes by estimating the human exposure to the hazard (e.g., pathogen) via each route. For each known transmission route, this approach requires information on the prevalence and/or dose/concentration of the pathogen in the source, the changes of the prevalence and quantity of the pathogen throughout the transmission chain, and the frequency at which humans are exposed by that route (e.g., consumption data). Exposure doses are then compared, and the relative contribution of each of the various transmission routes to human exposure in the population is estimated, proportionally to the size of each exposure dose.

The data requirements of the comparative exposure assessment approach depend on the overall transmission groups considered in the model (i.e., foodborne, environmental, and/or contact with animals), as well as on the point in the transmission chain where the “origin” of the pathogen is set. In general, contamination data for each source, information on the main steps in the transmission chain and data on the effects of these on contamination, and exposure data are needed. If transmission via contact with live animals is considered, the exposure model needs to be expanded to consider different possibilities for direct and indirect contact with a contaminated animal.

Exposure assessments have been used with different degrees of success to source-attribute disease by several microbial agents (namely, *Listeria*, *Campylobacter*, Verocytotoxigenic *Escherichia coli* (VTEC), and *Toxoplasma gondii*) and by chemical hazards (aflatoxins, cadmium, and lead) (27–34).

In the context of AMR, this approach is particularly useful to address a widely recognized knowledge gap, which is understanding the relative contribution of the exposure routes of AMR from animals to humans. Specifically, it can be used to estimate the relative importance of the food chain, companion animals, and the environment for exposure of the general population to antimicrobial-resistant bacteria or AMR genes. Thus far (and to our knowledge), two comparative exposure assessments have been applied to estimate the relative contribution of different types of meat to the exposure of consumers to extended-spectrum beta-lactamase-

producing and AmpC beta-lactamase-producing *E. coli* in the Netherlands (17) and in Denmark (35).

An important drawback of this approach is that, due to data limitations and gaps (e.g., in food preparation habits and the effect of these on the contamination of foods), exposure estimates for microbial pathogens are likely to present wide uncertainty intervals. Furthermore, in the context of AMR, these studies focus on specific antimicrobial-resistant pathogens and do not address all concomitant transmission routes contributing to overall transmission of resistance to humans (e.g., the same AMR determinant present in other members of the meat bacterial community), which adds to the uncertainty of the relative exposure estimates.

Epidemiological approaches

Epidemiological approaches for source attribution include analyses of data from outbreak investigations and studies of sporadic infections; both approaches attribute illness at the point of exposure. An outbreak is here defined as (i) the occurrence of two or more cases of a similar illness resulting from the exposure to a common source (36) or (ii) a situation in which the observed number of cases exceeds the expected number and where the cases are linked to the same food source. Sporadic cases are those that have not been associated with known outbreaks (37). Even though outbreak-associated cases are more likely to be captured by public health surveillance systems, an unknown proportion of cases classified as sporadic may be part of undetected outbreaks.

Many outbreak investigations are successful in identifying the specific contaminated source or ingredient causing human infections. A simple descriptive analysis or summary of outbreak investigations is useful for quantifying the relative contribution of different foods to outbreak illnesses. However, these implicated foods may be composed of multiple ingredients, and thus outbreak data do not always allow pinpointing of the actual source of infection. Probabilistic models using outbreak data to estimate the total number of illnesses in the population attributable to different foods provide a useful way to generalize outbreak data to a broader population of foodborne illnesses. These models are used not only to generalize the results of outbreak investigations, but also to estimate the contaminated sources in composite or “complex” foods.

Analyses of data from outbreak investigations benefit from detailed data on each reported outbreak and require the adoption of a food categorization scheme for classification of implicated foods (see, e.g., 38). Composite foods are assigned to two or more food categories

depending on the number and nature of their ingredients. By assigning a probability to each ingredient, corresponding to the likelihood that it was the source of the outbreak, outbreak data, including data about both simple and complex foods, can be used to attribute foodborne illnesses to sources.

Several analyses of outbreak data for source attribution have been published in recent years, most of them modeling (39–41) or summarizing (42, 43) data from multiple pathogens. The strength of this method is that it uses data that is readily available in many countries worldwide, and thus its use is not restricted to countries with integrated foodborne disease surveillance programs. Also, it attributes foodborne illnesses at the point of exposure, which means that it is particularly useful to identify which foods (including processed foods) most frequently cause disease, as well as which risk factors contribute more to contamination of foods at the end of the food chain (e.g., cross-contamination). This type of information is valuable to define interventions at the processing and consumption level but does not provide evidence to inform risk management strategies at the origin of the pathogen (reservoirs).

Several outbreaks caused by antimicrobial-resistant pathogens have been reported and investigated in the past decades (see, e.g., 44, 45). A review of outbreak data has also been used for source attribution of antimicrobial-resistant *Salmonella* in the United States, suggesting that antimicrobial susceptibility data on isolates from foodborne outbreaks can help determine which foods are associated with resistant infections (46). Even though few countries or regions are likely to have sufficient data for a robust source attribution analysis using AMR-related outbreaks, summarizing available information may provide evidence on the relative contribution of different foods to infection with antimicrobial-resistant pathogens.

Another epidemiological approach that can be used for source attribution of foodborne disease is the case control study of sporadic cases. Case-control studies are a valuable tool to identify potential risk factors for human illness, including sources and predisposing, behavioral, or seasonal factors (47). In addition to individual case-control studies, a systematic review of published case-control studies of sporadic infections of a given pathogen can provide an overview of the relevant exposures and risk factors for that disease and a summary of the estimated population-attributable fractions for each exposure (48). A systematic review follows a rigorous search strategy to identify all potentially relevant peer-reviewed case-control studies for a hazard,

studies being conducted in a variety of countries and time periods, designed with different settings, and potentially focused on specific age groups within the population. A meta-analysis is then performed to compare and combine information from different studies. To do this, risk factors may be stratified according to source-categorization schemes, location of exposures, and if appropriate, frequency of exposure. An overall population-attributable fraction derived from a meta-analysis or weighted summary of several case-control studies of a certain hazard can be combined with estimates of the burden of disease caused by that hazard to estimate the burden of disease attributed to each exposure.

This method is particularly useful for hazards that do not frequently cause outbreaks but that have been extensively studied (49). In addition, it is valuable to attribute illness at a regional or global level when data are scarce in most countries. A number of case-control studies have been conducted to investigate risk factors for infection with foodborne pathogens resistant to antimicrobials (see, e.g., 50, 51). However, the utility of a meta-analysis of case-control studies to investigate the relative contribution of different sources and risk factors to infection with antimicrobial-resistant pathogens may be limited if a low number of case-control studies focused on specific antimicrobial-resistant pathogens or AMR genes have been conducted.

Other approaches

Other approaches for source attribution of foodborne pathogens include intervention studies and expert elicitations. Intervention studies are large-scale, well-structured prospective studies that are specifically tailored to evaluate direct impacts of a specific intervention on the risk of disease in a population. While they would be the “gold standard” of attribution studies, they have the disadvantages of being resource-demanding, expensive, and difficult to implement because other concurrent factors may affect occurrence of disease.

Expert elicitation can be designed as structured methods to gather and analyze knowledge from experts, which is communicated with a measure of uncertainty. They are particularly useful to attribute the burden of foodborne diseases to the main transmission pathways (i.e., foodborne, environmental, direct contact), for which data-driven methods are typically insufficient (49). There are numerous methods used for expert elicitation, including methods that are based on iteration and finding consensus among a small group of experts (e.g., the Delphi method). Expert judgments are subjective by nature and may be biased by the specific back-

ground and scientific expertise of the respondents, and several methods to evaluate the experts’ performance have been described. Several expert elicitation studies have been conducted for source attribution of foodborne disease (e.g., 52, 53). The World Health Organization’s Initiative to Estimate the Global Burden of Foodborne Diseases (WHO-FERG) has undertaken a large-scale and successful expert elicitation to attribute disease due to 19 foodborne hazards to main transmission groups at global, regional, and subregional levels (54). The study applied structured expert judgment using Cooke’s classical model (55) to obtain estimates for the relative contributions of different transmission pathways for several foodborne hazards.

Applications and Results

Despite the increased recognition of the importance of source attribution of foodborne pathogens to direct risk management strategies, and the growing use of these approaches in several countries and research groups, source attribution of AMR is still in its infancy. There are few published examples of the methods described here, and the identified challenges are still being addressed. The two microbial subtyping studies that have been published are both frequency-matched studies that used antimicrobial susceptibility patterns as a typing method for *Salmonella* (15, 16). These studies use AMR profiles as a typing method (i.e., to characterize pathogen subtypes) but do not focus on the source origin of specific AMR genes. Still, they are able to estimate the distribution of AMR in human cases attributed to different sources, as is done routinely in the *Salmonella* source attribution activities in Denmark (56). Similarly, the two comparative exposure assessments that have been applied to estimate the relative contribution of different types of meat to the exposure of consumers to AMR have focused on the same causative agent, extended spectrum beta-lactamase-producing and AmpC beta-lactamase-producing *E. coli* (17, 35). These studies demonstrate that the method could be extended to other countries and agents. The recent review of outbreak data for source attribution of antimicrobial-resistant *Salmonella* in the United States suggests that antimicrobial susceptibility data on isolates from foodborne outbreaks can help determine which foods are associated with resistant infections (46). This method could be applied in countries that have sufficient data or to regional data in an attempt to gather information from multiple countries. Numerous epidemiological studies of sporadic infections (case-control or cohort studies) investigating risk factors for antimicrobial-resistant infections in humans

demonstrate these methods' usefulness to identify routes of AMR (e.g., [57–59](#)). While their use focusing on food-borne or direct or indirect contact to animals' transmission has been limited, available studies still provide information that is useful for food safety risk management ([50, 51](#)).

Strengths and Weaknesses

Source attribution of AMR genes and of antimicrobial-resistant pathogens is a research area under active development. The application of the methods described here remains a challenge, for reasons that depend on each method considered.

For the application of subtyping frequency-matched studies, two of the main challenges are the limited availability of animal, food, and human AMR data from established surveillance systems and the difficulty of defining the number of AMR profiles highly specific to a particular source/transmission route, a cornerstone of this method. Furthermore, the fact that the method does not determine the actual transmission route from each specific reservoir to humans is another limitation for the use of frequency-matched models. Due to the public health need for understanding the transmission of AMR, population genetics approaches may eventually be a good complement to frequency-matched models, especially considering the increasing availability of whole-genome sequencing and metagenomics data, which describe the occurrence of AMR genes in populations. For instance, population genetics can help in identifying reservoir-specific AMR genes' patterns that can then be used in frequency-matched models. New-generation sequencing data may also contribute to unraveling details that contribute to a more accurate source attribution, such as the evolution of AMR patterns over time in different sources and resistance in humans that is not transmitted from animals or foods.

While single genomics and metagenomics studies may support the development of novel subtyping source-attribution methods, they may hinder the application of comparative exposure assessment. Information on the prevalence and quantity of AMR genes or antimicrobial-resistant pathogens in each source, as well as their changes throughout the transmission chain, are difficult to assess from those data and are impaired by a high degree of uncertainty.

Epidemiological methods of source attribution, e.g., those based on outbreak investigation, have the advantage of not relying on a sophisticated, data-abundant, and integrated surveillance system encompassing animal reservoirs, foods, and humans. However, they require

consistent AMR investigation on food sources and human cases, based at least on bacterial isolation and phenotypic susceptibility testing. Eventually, new-generation sequencing may overtake traditional diagnostic methods in outbreak investigation ([14, 60](#)), which will also require modification of the current epidemiological approaches.

Intervention studies have, in the context of AMR, the same limitations as when applied to bacterial pathogens. It is difficult to evaluate the exact impact of a specific intervention (e.g., reducing antimicrobial use at the farm level) on the population where disease is attributed (e.g., AMR occurrence in humans). Control measures that reduced antimicrobial use in primary production have been successfully implemented with the aim of reducing AMR in animals (e.g., the antimicrobial growth promoter intervention, the voluntary ban on the use of cephalosporins, and the yellow card antimicrobial scheme in swine herds in Denmark [[61–63](#)]). However, to assess the real success of such measures in terms of public health impact, it is necessary to collect data prior to and following the intervention ([14](#)), at all dimensions of AMR transmission to humans, i.e., also including other transmission routes such as the environment and antimicrobial use in humans.

RISK ASSESSMENT

Microbial risk assessment of AMR

Risk assessment is the process of estimating the likelihood that exposure to a biological, chemical, or physical hazard will result in an adverse health effect in exposed individuals. Microbial risk assessment has been established as a part of the food safety risk analysis paradigm by international and national bodies in recent decades, with harmonized guidelines being proposed and widely adopted worldwide ([8, 64](#)). In the context of AMR, risk assessments are useful to inform regulatory decision-making for the mitigation of potential health consequences in both humans and animals ([65](#)). While the importance and need for AMR risk assessments have been recognized for decades ([66](#)), their application has been complicated by several knowledge gaps. Challenges of the development of AMR risk assessment include the following:

- The nature of the hazard is difficult to identify and will determine the nature of the adverse consequence of the exposure. In the context of AMR risk assessment, different hazards can be considered ([67, 68](#)). For example, Salisbury et al. ([67](#)) discussed three interrelated hazards that can be

assessed separately: the antimicrobial drug, the antimicrobial-resistant bacteria, and the AMR determinant, leading to three different health consequences, respectively—development of resistance, infection and treatment failure, and transference of resistance. Similarly, Manaia (68) describes that resistome-associated risks have been discussed considering the microbial community, the genome, and transmission of resistance.

- The nature of the risk posed by antimicrobial use and AMR to human health is inherently complex and logically linked to the nature of the hazard, as mentioned above. In other words, while the likelihood that humans will be infected by pathogens that are resistant to one or several antimicrobials can be estimated, the resulting adverse health consequences can be one or several of the following: development of disease due to infection with the pathogen, failure of treatment of the infection due to resistance to the drug(s) used, and spread of AMR genes to commensal bacteria in the human host (which can amplify the risk and extend the impact of an isolated exposure in time).
- Numerous factors are involved in the process of selection and spread of resistance in bacterial populations, between and within animal species, humans, and the environment and within different bacterial populations in those same reservoirs. These factors include the several drivers of the emergence and spread of AMR in food production, specifically at the farm. At this level, antimicrobial use is recognized as the most important driver but is not always necessary (if, for example, coresistance and coselection occur) and not always sufficient; additional drivers are, e.g., poor prevention and control of infectious diseases leading to increased antimicrobial use and the spread of clones that have established themselves in the herd/environment and keep selective pressure, even if antimicrobial use is interrupted. These factors, among many others, influence the development of exposure assessment in microbial risk assessment.
- In addition to the challenges described above, estimating the likelihood of adverse health effects, given exposure to an antimicrobial-resistant pathogen or determinant, is difficult due to the absence of a well-defined dose-response effect for AMR and the existence of various possibilities of adverse effects.

Recognizing the need for AMR risk assessments to identify strategies aimed at preventing and reducing

the disease burden of AMR transmitted through foods, a number of reviews and scientific articles proposed frameworks for such risk assessments in the late 1990s and early 2000s (66, 67, 69). Even though such proposals were comprehensive and structured to address the challenges identified at that time, they were not widely adopted, mostly due to remaining knowledge and data gaps regarding AMR transmission and impact. More recent frameworks apply current available data and are either mostly qualitative or semiquantitative (see, e.g., 70, 71), take a linear approach (e.g., 72), and/or focus on marketing authorization applications for antimicrobial veterinary medicinal products for use in food-producing species (73).

The Four Steps of Microbial Risk Assessment Focusing on AMR

The microbial risk assessment process is, as described by the *Codex Alimentarius* guidelines (8), constituted by four main components: hazard identification, hazard characterization, exposure assessment, and risk characterization.

In an AMR risk assessment, the hazard can be the antimicrobial drug, the antimicrobial-resistant pathogen, or the AMR determinant. Ultimately, the identification of the hazard of interest depends on the risk-assessment question to be addressed. In a traditional microbial risk assessment (i.e., focused on a pathogen-food pair, without considering resistance to antimicrobial drugs), the hazard identification step consists of the qualitative description of the hazard, including the evaluation of the presence of the pathogen in a food product available for consumption in a population and the host interface (types of disease caused, susceptible populations). In the context of AMR, this step is complicated by a number of factors: (i) selection of resistance in a pathogen can occur by multiple mechanisms (namely, mutation and horizontal gene transfer of mobile genetic elements containing AMR genes) (74); (ii) one or more genes may be necessary for the development of AMR; (iii) AMR genes can be located in chromosomal or extra-chromosomal DNA, such as plasmids (74); and (iv) several bacterial species or strains can harbor and serve as a reservoir for resistance.

The hazard characterization step of a risk assessment consists of the review and collection of information on the relationship between the dose of the hazard and the onset of disease in the exposed individuals (i.e., infectious dose) and the relationship between different doses and the probability of occurrence of disease (i.e., dose-response). The response of a human population to

exposure to a foodborne pathogen is highly variable, reflecting the fact that the incidence of disease is dependent on a variety of factors such as the virulence characteristics of the pathogen, the numbers of cells ingested, the general health and immune status of the hosts, and the attributes of the food that alter microbial-host interactions (75). Thus, the likelihood that any individual becomes ill due to an exposure to a foodborne pathogen is dependent on the integration of host, pathogen, and food matrix effects. Again, in AMR risk assessment, the required data to assess a dose-response relationship depend on the hazard considered; it can be one of the three: the dose level of the antimicrobial for observing resistance usually expressed by the MIC breakpoint (74), or any other factor that can affect the development or amplification of resistance, the dose of the pathogen needed to cause disease, or any factor related to the stability and transfer potential of the AMR gene in a bacterial population (67).

In the exposure assessment step, the likelihood that an individual or a population will be exposed to a hazard and the numbers of the microorganism that are likely to be ingested are estimated (76). The exposure assessment requires data on the prevalence and concentration of the hazard in the food source(s), as well as information on the potential changes of the pathogen load throughout the food-processing chain (e.g., growth, reduction) (77); in addition, it requires data on the frequency and amount of food consumed by individuals of the population. As mentioned above, numerous factors influence the process of selection and spread of resistance, consequently influencing the final exposure of the consumer to AMR genes or antimicrobial-resistant pathogens. These factors are either still unknown or there are limited data reporting their influence on AMR transmission throughout the food chain.

In the last component of a risk assessment, risk characterization, the final risk to the consumer is estimated by integrating the previous three components. Specifically, the measure of exposure (i.e., the likely dose an individual is exposed to in a given food consumption/exposure event) is integrated with the dose-response relationship to estimate the likelihood of an adverse health effect. In the context of AMR microbial risk assessment, even after an appropriate definition of the risk question and the targeted hazard identification (which determine the adverse effect to be assessed) and the estimation of the likelihood of exposure to the hazard of interest, characterizing the risk in the absence of an appropriate and comprehensive hazard characterization step remains a challenge. A “dose-response” step

becomes particularly demanding when the “dose” at exposure is expressed in genotypic terms (by use of genomics or metagenomics AMR data) and the “response” must be expressed in phenotypic terms (e.g., expression of resistance in a pathogen or horizontal transfer of an AMR gene between commensal and pathogenic bacteria).

Applications and Results

A number of risk assessments focused on specific antimicrobial-resistant pathogen-food/animal pairs have been conducted since the publication of the various proposed guidelines. These include qualitative, semi-quantitative, and quantitative risk assessments performed by food authorities, academia, or industry. Here, we provide examples of the three types of risk assessment that have been important to highlight the challenges and limitations they still face, the applications of their results, and the need for further studies.

Qualitative risk assessments

One of the first studies published assessed the health impact of residues of antibacterial and antiparasitic drugs in foods of animal origin and was published over 2 decades ago (78). It was a qualitative and comprehensive review that focused on residues of a variety of drugs in multiple foods, and it was an important step for the recognition of several of the challenges described in this chapter. More drug- and pathogen-focused qualitative assessments have been conducted since then, including in recent years, such as the qualitative risk assessment focused on methicillin-resistant *Staphylococcus aureus* (MRSA) conducted by a multisectorial and interdisciplinary expert group in Denmark (79). This study is a good example of an applied risk assessment, conducted upon request from the food and veterinary authorities with the aims of (i) assessing the risk of livestock MRSA based on the existing knowledge and the results of veterinary screening studies conducted in herds and (ii) providing a recommendation for control measures to reduce the spread of MRSA from the affected herds to the surrounding environment and community. The method consisted of a comprehensive evaluation of all available data on the prevalence of MRSA in animals and humans, as well as on the risk factors for infection by livestock MRSA from the environment, from meat, from occupational activities (e.g., risk for slaughterhouse or farm workers), and from the community. The risk assessment consisted of a descriptive evaluation of the risk of these types of transmission in the Danish population.

Another recent study has applied the risk assessment framework developed by the European Medicines Agency (73) to assess the AMR risk to public health due to use of antimicrobials in pigs, using pleuromutilins as an example (80). Livestock-associated MRSA of clonal complex 398 (MRSA CC398) and enterococci were identified as relevant hazards. This framework followed the International Organization for Animal Health's approach to risk assessment and consisted of four steps describing the risk pathway, combined into a risk estimate. The study applied a qualitative approach, where the output of each step was defined on a scale. Likewise, the level of uncertainty was described qualitatively in the different steps and the output (as high, medium, or low). The authors discuss the value of mathematical modeling as a tool to simulate pathways and identify ways of reducing resistance. Still, they stress that the relationship between reducing consumption of antibiotics and reducing resistance is not necessarily linear and defend that this relationship needs to be better established for modeling to have full value (80). Despite the fact that this study is recent at the time of writing of this chapter and thus could build on all newly available evidence on AMR mechanisms, it still dealt with substantial data and knowledge gaps that enhanced uncertainty around outputs (80).

Another example of a qualitative assessment is the WHO's list of critically important antimicrobials (70). The list applies criteria to rank antimicrobials according to their relative importance in human medicine. The purpose of this assessment is to provide clinicians, regulatory agencies, policy-makers, and other stakeholders' with information to develop risk management strategies for the use of antimicrobials in food-production animals globally. The first WHO list of critically important antimicrobials was developed in a WHO expert meeting in 2005, where participants considered the list of all antimicrobial classes used in human medicine and categorized them into the three groups "critically important," "highly important," and "important" based on two criteria that describe, first, the availability or lack of availability of alternatives to the antimicrobial for treatment of serious bacterial infections in people, and second, if the antimicrobial is used to treat infections by (i) bacteria that may be transmitted to humans from nonhuman sources or (ii) bacteria that may acquire resistance genes from nonhuman sources. The output of the qualitative assessment is a list of classes of drugs that meet all three of a set of defined priorities. Since its original publication, the assessment has been revised several times and is now in its fifth edition.

Semiquantitative risk assessments

One example of a semiquantitative assessment is the study integrating a probabilistic quantitative risk assessment conducted in Denmark to assess the human health risk of macrolide-resistant *Campylobacter* infection associated with the use of macrolides in Danish pig production (81). This model was able to account for exposure through imported and domestic meat (i.e., that could be a vehicle for antimicrobial-resistant bacteria as a consequence of antimicrobial drug use in animal production in that country) and used evidence available at the time. One important feature of this study is that, while it measured exposure probabilistically and thus reflected model and data uncertainty, the final step of the risk assessment—risk characterization—used an ordinal scale, and thus risk was described in a qualitative scale.

Quantitative risk assessments

Several quantitative risk assessments have been published since the early 2000s. These include the high-profile assessment of fluoroquinolone-resistant *Campylobacter* from chicken in the United States (82), which ultimately prompted the Food and Drug Administration to propose withdrawal of the approval of new animal drug applications for fluoroquinolone use in poultry, an action that would prohibit fluoroquinolone use in chickens and turkeys in that country (83).

Another early study employed probabilistic methodology to analyze the potential public health risk from *Campylobacter jejuni* and fluoroquinolone-resistant *C. jejuni* due to fresh beef and ground beef consumption (84). The model focused on the beef product at retail and modelled consumer handling in the kitchen, processing, and consumption. The model estimated, first, the risk of *Campylobacter* infection through consumption of beef and then the risk of treatment failure given infection, concluding an increased health impact due to resistance.

In another study, a risk assessment followed the Food and Drug Administration's Center for Veterinary Medicine Guidance (85) and was commissioned by a pharmaceutical company to estimate the risk of human infection treatment failure associated with the use of an antimicrobial drug in food animals (86). The deterministic model included all uses of two macrolides in poultry, swine, and beef cattle. The hazard was defined as illness (i) caused by foodborne bacteria with a resistance determinant, (ii) attributed to a specified animal-derived meat commodity, and (iii) treated with a human use drug of the same class. Risk was defined as the probability of this hazard combined with the consequence of treatment failure due to resistant *Campylobacter* spp.

or *Enterococcus faecium*. At the time, this microbial risk assessment had the advantage of being quantitative and thus more transparent when compared to previous assessments focusing on AMR. Thus, the authors highlighted several limitations, particularly with regard to data gaps on the probability of treatment failure due to the antimicrobial-resistant bacteria and the probability of resistant determinant development. In contrast to many evidence and risk assessments conducted elsewhere, the results of this study led the authors to conclude that the current use of macrolides in cattle, poultry, and swine created a risk much lower than the potential benefit to food safety, animal welfare, and public health (86).

The same author published another risk assessment a few years later, applying a similar approach to estimate the risk of a different combination of antimicrobial-pathogen-fluoroquinolone-resistant *Salmonella* and *Campylobacter* in beef in the United States (87). This approach was able to provide a better measure of uncertainty but was similar in its findings, concluding that the risk of health consequences in humans was minimal.

The most recent quantitative risk assessment study published is also the more novel and promising of the AMR studies reviewed here (88). It considered the existence of environmental compartments resulting from sewage-treatment plants, agriculture production, and manufacturing industries and assessed their roles in the maintenance, emergence, and possible dissemination of antibiotic resistance. This study used probabilistic methods to assess the risks of antibiotic resistance development and neutralizing antibiotic pressures in hotspot environments. Importantly, this study presents a modeling approach to assess the selective pressure exerted by antibiotics in bacterial communities and to calculate antibiotic resistance development risks. While the described approach was exemplarily when used to model antibiotic resistance risks in an intensive aquaculture production scenario of Southeast Asia, it has the potential to be applied to other cases, including other types of animal production, settings, and drugs.

Strengths and Weaknesses

Microbial risk assessment is a science-based tool with proven benefits in supporting food safety authorities in policy-making. It is hence commended to continue its use in assessing the consequences for the consumer of the transmission of AMR genes/pathogens through the food chain. The fact that it is a well-defined, stepwise-structured method facilitates its adaptation to the food safety challenge of AMR. However, several limitations

have already been identified and require the joint focus of the scientific community, risk assessors, and authorities. Examples of a few critical challenges are the following:

- The definition of AMR is critical for the four steps of microbial risk assessment and needs, therefore, to be well established at the start of a risk assessment study. Martínez et al. (74) explains the existence of several possible definitions of resistance (namely, clinical, epidemiological, and operational) and two definitions of resistance gene (ecological and operational). The adoption of standard concepts and terminology is a requirement for the transparency of microbial risk assessment and an important part of its development. Although transmission of AMR genes and antimicrobial-resistant bacteria may be perceived and have been defined as two separate hazards, it has also been recently suggested that the risk of AMR transmission to humans cannot be estimated unless the AMR gene pool and the presence and quantity of antimicrobial-resistant bacteria that are able to colonize and multiply in the human body are both taken into consideration (68).
- Exposure assessment often relies on available knowledge of the changes in the microbial hazard levels throughout the food chain, due to, e.g., growth or inactivation. In the context of microbiomes and resistomes, it is difficult to model these changes, because the very composition of the microbial population (and corresponding AMR genes) may significantly change between “farm” and “fork” (89, 90). Consequently, microbial risk assessment for AMR is highly dependent on data collected at several points of the transmission pathway, both from the source(s) of AMR and from exposed human subjects.
- While new-generation sequencing attractively provides a broad characterization of the presence and abundance of AMR genes in a particular pathogen or in the microbiome from a particular reservoir, it remains a challenge to determine variability of the resistome and of the potential to exchange AMR genes (i.e., presence of phage recombination sites, plasmids, integrons, or transposons) between different pathogen strains (68). This knowledge is crucial to assign the AMR genes detected with metagenomics to the corresponding bacterial hosts and to account for the occurrence of horizontal gene transfer between commensal and pathogenic bacteria in a population.

- An important challenge for the integration of metagenomics data in microbial risk assessment is the harmonization of languages between the “omics” and the food microbiology communities (91).
- Risk characterization requires knowledge of the relationship between a “dose,” resulting from the exposure assessment, and a “response,” i.e., the adverse health effect of exposure. However, the infective dose and the modes of transmission of most of the antimicrobial-resistant bacteria of relevance are still unknown (68), which represents an important knowledge gap for the development of microbial risk assessment for AMR.
- Finally, a major limitation of the current microbial risk assessment frameworks is that they do not allow estimation of the long-term impact of exposure to AMR. Particularly serious public health consequences of AMR arise when multiresistant bacteria emerge and become widespread. There is therefore the need to develop microbial risk assessment methods that include a different characterization of the risk of AMR. In addition to immediate consequences to human health due to a single exposure to an antimicrobial-resistant pathogen, it is necessary to estimate the likelihood that such exposure (eventually together with past and subsequent ones, to the same or other types of AMR) will lead to the development of antimicrobial multiresistance in the future. Also, it is necessary to assess the potential of multiresistance spread, to characterize the severity of the consequences of exposure to multiresistance, and to estimate the time from initial exposure to those consequences.

DISCUSSION AND FUTURE PERSPECTIVES

Several position and stakeholder papers have stressed the need for improved quality and an increased amount of data for risk assessment of AMR (see, e.g., 92). These include, e.g., data on antimicrobial use in animal production; AMR surveillance data in animals, foods, and humans; and gene transfer and spread of AMR genes. All data requirements apply for most source attribution studies, and thus are transversal to the methods described in this article. Likewise, many of the challenges to the application of these methods in the context of AMR are common to source attribution and risk assessment approaches (Table 1).

The studies described here all show the importance of knowledge of (i) the most important sources and routes of transmission of antimicrobial-resistant bacteria or AMR genes, (ii) the actual risk for human health, and (iii) the points in the transmission chain where interventions could be effective to reduce this risk. While all findings so far have been crucial to direct policies and raise awareness to the public health impact of AMR in animals and foods, they are insufficient for a complete understanding of the underlying transmission mechanisms and the real impact of AMR. Several challenges have been addressed, including the fact that emergence and spread of AMR is complex. From an epidemiological point of view, the risk of AMR probably follows the “sufficient-component causes” principle (93). The sufficient-component causes principle is an epidemiological causal modeling approach that can be used to explain diseases, or conditions like AMR, characterized by many causes, none of which alone is necessary or sufficient. The relations among the causes are described such that a

TABLE 1 Definition, overview of methods and main challenges of source attribution and microbial risk assessment approaches

	Source attribution	Microbial risk assessment
Definition	Partitioning of human cases of illness to the responsible sources (e.g., foods, animal reservoirs)	Systematic and science-based approach to estimate the risk of microbial hazards in the production-to-consumption chain
Methods	Microbial subtyping Comparative exposure assessment Outbreak-data analysis Case-control studies Expert elicitations Intervention studies	Qualitative RA ^a Semiquantitative RA Quantitative RA Deterministic Probabilistic
Main challenges in the context of AMR	Hazard identification, e.g., the antimicrobial drug, the antimicrobial-resistant pathogen, or the AMR determinant Lack of occurrence/prevalence data Definition of the health outcome, i.e., infection with antimicrobial-resistant agent, treatment failure (in case treatment is needed), or spread of resistance determinant between commensal and pathogenic organisms Lack of epidemiological data	Establishment of dose-response relationship Determining variability of the resistome and of the potential to exchange AMR genes between different pathogen strains

^aRA, risk assessment.

“sufficient cause” is a set of minimal conditions that will definitely lead to the outcome (e.g., antimicrobial-resistant infection), and a “component cause” is one of the minimal conditions included in a sufficient cause (93). For example, a particular resistance gene can be a component cause of an antimicrobial-resistant infection, but the sufficient cause of the latter includes other conditions, such as the bacterial strain carrying that particular gene, that pathogen causing infection, treatment of the infection with antimicrobial(s) for which resistance is encoded in the gene, and the actual expression of that resistance gene. The future of microbial risk assessment for AMR may therefore include defining the components sufficient to cause AMR transmission from animals/foods/the environment to humans followed by treatment failure of infections by antimicrobial-resistant pathogens.

Recent developments in omics technologies (whole-genome sequencing and metagenomics, transcriptomics, proteomics, metabolomics, fluxomics) provide unique opportunities to fill in some of our knowledge gaps. It is now widely recognized that these omics technologies have advantages compared to traditional phenotypic culture-based methods for characterizing microorganisms (91, 94).

Brul et al. (91) described in detail how omics can be integrated in each step of microbial risk assessment, contributing to a mechanistic insight into the interaction between microorganisms and their hosts, new perspectives on strain diversity and variability and physiological uncertainty, and overall more robust risk assessments. den Besten et al. (94) discussed the utility of omics technologies applied by the food industry, to help identify the influence of different bacterial ecosystems on both pathogen survival and growth—information that can eventually contribute to the future definition of food safety objectives.

A particular advantage of metagenomics is that it provides a picture of the whole microbial community and its resistome, which is key to understanding AMR emergence and spread in a population. Importantly, these new “typing” techniques have been rapidly followed by new bioinformatics and new statistics/modeling tools that allow for the analysis and sense-making of such (big) data (91, 95). For example, machine learning has the potential to be applied on the analysis of omics data. Combining machine learning approaches with metagenomics and farm-specific data could allow for describing, e.g., health, production efficiency, and the relative abundance of AMR genes, based on the identification of (clusters of) genetic factors in the farm

microbiome. In addition, such techniques could be used to examine the predictive importance of (clusters of) genetic factors to characterize (i) a “healthy farm microbiome” or (ii) AMR genes in a specific animal reservoir. They can also be used to identify (combinations of) specific husbandry practices that are associated with, e.g., a particular resistome or a healthy farm microbiome. The latter could lead to recommendations on how to shift the farm microbiome to improve the overall health of the farm and consequently, in the long term, to reduce the level of antimicrobial use and antimicrobial-resistant bacteria. It is possible that promoting a healthy farm microbiome will have a more long-term impact on the overall reduction of AMR than focusing exclusively on the farm resistome. Metagenomics and other omics technologies hence have enormous potential for the future development of source attribution and microbial risk assessment of AMR through foods. To explore their full potential, different technologies will be combined. For example, genomics studies should be coupled with proteomics, because gene-expression studies do not always reflect the actual protein levels (91). Also, genomic similarities may not imply similarities in behavior, because the surrounding environment (food matrix, bacterial ecosystem, etc.) also plays a role (94). Furthermore, omics data are not sufficient without accompanying epidemiological data that allow for the identification of risk factors for AMR.

CONCLUDING REMARKS

Recent developments in source-attribution and microbial risk assessment of AMR are promising and have significantly contributed to the evolution of each of these methods. However, the adaptation to the omics big data era is happening at a much slower pace than the speed at which these data are becoming available. This is due to the many challenges encountered when interpreting those data.

AMR at the animal reservoir, food, environment, and human levels is increasingly described by the characterization of the resistomes of single bacteria isolates (by whole-genome sequencing) or the bacterial whole community (by metagenomics) representing each of those populations. Gradually, AMR surveillance will convert from phenotypic to genotypic (e.g., PulseNet International is already on its way to standardizing whole-genome sequencing-based subtyping of foodborne disease [95]). For a successful transition, it is crucial to pair genomic data with phenotypic data and relevant explanatory epidemiological data.

This transition will require a parallel adaptation of the existing analysis methods, which will include the development of new source-attribution and microbial risk assessment modeling approaches. It is therefore with great expectation that we foresee in the near future a surge of influencing and inspiring scientific output in both fields.

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